One-Hour-Ahead Bitcoin Volatility Forecasting with Multi-Horizon Volatility, Market Activity, and Time-Varying Parameters

Ding Minzhao

July 2026

Abstract

This study evaluates one-hour-ahead Bitcoin volatility forecasts using five-minute BTCUSDT USD-margined perpetual futures data from January 2022 to December 2025. Hourly realized variance is forecast using four models: a persistence benchmark, an AR(24) model, a fixed-parameter HAR-X model, and a time-varying parameter HAR-X state-space model. Forecast performance is assessed through a chronological training–validation–test design using MAE, RMSE, and QLIKE.

On the 2025 test sample, Fixed HAR-X achieves the lowest QLIKE, while TVP-HAR-X records the lowest MAE and RMSE. The results show that multi-horizon volatility components and trading activity improve forecast accuracy relative to both persistence and AR(24). Time-varying coefficients provide modest gains in log-scale forecast accuracy, whereas the fixed HAR-X specification performs best under variance-scale loss. The value of parameter adaptation therefore depends on the forecasting objective and evaluation criterion.

Contents

1 Introduction **4**

2 Data and Variable Construction **5**
2.1 Data Source and Sample Period 5
2.2 Five-Minute Returns 5
2.3 Hourly Realized Variance 5
2.4 Data Quality Control 6
2.5 Multi-Horizon Volatility Features 6
2.6 Downside Realized Semivariance 6
2.7 Trading Activity 7
2.8 Aggressive-Buying Proxy 7
2.9 Feature Standardization 7
2.10 Final Modeling Sample 7

3 Forecasting Design **8**
3.1 One-Hour-Ahead Forecasting Problem 8
3.2 Chronological Sample Division 8
3.3 Common Evaluation Sample 8
3.4 Sequential Forecasting and Information Control 8
3.5 Model Selection Rules 9
3.6 Mean Absolute Error 9
3.7 Root Mean Squared Error 9
3.8 QLIKE Loss 9
3.9 Forecast Loss Comparison 10

4 Methodology **10**
4.1 Modeling Strategy . . . 10
4.2 Persistence Benchmark . . . 11
4.3 Fixed-Parameter Autoregressive Model . . . 11
4.3.1 Stationarity Assessment . . . 12
4.3.2 Order Selection and Estimation . . . 12
4.4 The HAR-X Specification . . . 12
4.5 Fixed HAR-X Model . . . 13
4.6 Motivation for Time-Varying Parameters . . . 14
4.7 TVP-HAR-X State-Space Model . . . 14
4.8 Kalman Filtering and Sequential Forecasting . . . 14
4.9 Missing Observations . . . 15
4.10 Parameter Estimation . . . 16
4.11 Interpretation of the Time-Varying Coefficients . . . 16
4.12 Model Diagnostics . . . 16
4.13 Final Model Comparison . . . 17

5 Empirical Results **17**
5.1 Stationarity and Dependence in Log Realized Variance . . . 17
5.2 Autoregressive Order Selection . . . 18
5.2.1 AR Stability and Residual Diagnostics . . . 18
5.3 Fixed HAR-X Estimation Results . . . 19
5.4 Fixed HAR-X Validation Performance . . . 19
5.5 TVP-HAR-X Parameter Estimates . . . 20
5.6 Time-Varying Coefficient Behavior . . . 20
5.7 TVP-HAR-X Validation Performance . . . 21
5.8 Final Out-of-Sample Test Results . . . 21
5.9 Predictive-Loss Significance Tests . . . 23
5.10 Validation–Test Comparison . . . 23
5.11 Summary of Empirical Findings . . . 24

6 Discussion **25**
6.1 Interpretation of the Main Result . . . 25
6.2 Validation and Test Performance . . . 25
6.3 Incremental Value of the HAR-X Information Set . . . 26
6.4 Interpretation of Trading Activity . . . 26
6.5 Downside Semivariance . . . 26
6.6 Aggressive-Buying Proxy . . . 26
6.7 Time-Varying Coefficient Patterns . . . 27
6.8 Stationarity and Parameter Variation . . . 27
6.9 Residual Dependence . . . 27
6.10 Statistical and Practical Significance . . . 28
6.11 Empirical Design . . . 28
6.12 Limitations . . . 28
6.12.1 Single Market and Contract . . . 28
6.12.2 Single Forecast Horizon . . . 28
6.12.3 Limited Microstructure Information . . . 28
6.12.4 Linear Predictor Effects . . . 29
6.12.5 Restricted State Covariance . . . 29
6.12.6 Retransformation from Log Variance . . . 29
6.12.7 Seasonality and Long-Memory Structure . . . 29
6.13 Directions for Further Research . . . 29

7 Conclusion **30**

Acknowledgements **31**

References **32**

A Reproducibility and Additional Results **33**

B Reproducibility **33**
B.1 Software Environment 33
B.2 Execution Procedure 33
B.3 Reproducibility Controls 33

1 Introduction

Bitcoin trades continuously and is characterized by pronounced and rapidly changing volatility. Reliable short-horizon volatility forecasts are relevant to risk management, position sizing, derivatives pricing, and intraday trading decisions. Previous studies show that high-frequency realized-volatility measures can support Bitcoin volatility forecasting, while jumps and structural breaks may materially affect predictive performance (Bergsli et al., 2022; Shen et al., 2020). These features motivate models that capture persistent volatility dynamics and allow forecasting relationships to evolve over time.

High-frequency returns provide a direct basis for measuring short-term price variation. Realized variance aggregates squared intraday returns and approximates the variation accumulated over a specified interval (Andersen et al., 2003). This study constructs hourly realized variance from five-minute returns and uses its logarithm as the forecasting target. The resulting measure retains substantially more intraday information than volatility proxies based on a single hourly or daily return.

Autoregressive models provide a natural benchmark because volatility is strongly persistent. However, a fixed collection of individual lags may represent multi-scale dependence inefficiently. The heterogeneous autoregressive model addresses this issue by combining realized-volatility components measured over different horizons (Corsi, 2009). The specification used here adapts that framework to hourly data through one-hour, six-hour, daily, and weekly volatility components.

The analysis also considers whether current market activity contains predictive information beyond lagged volatility. The mixture-of-distributions hypothesis links trading intensity and price variability through the arrival of information (Clark, 1973), while empirical evidence documents a close relationship between transaction counts and volatility (Jones et al., 1994). Three additional predictors are therefore included: downside realized semivariance, the number of trades, and an aggressive-buying proxy. Downside realized semivariance isolates the contribution of negative intraday returns to total realized variation (Barndorff-Nielsen et al., 2010).

A fixed-parameter model imposes a constant relationship between these predictors and future volatility. Such an assumption may be restrictive in cryptocurrency markets, where participation, liquidity, and volatility conditions can change substantially over time (Shen et al., 2020). To accommodate gradual parameter variation, this study estimates a time-varying parameter HAR-X model in state-space form. The coefficients follow random-walk state equations and are updated sequentially through the Kalman filter (Kalman, 1960; Durbin and Koopman, 2012).

The empirical analysis uses five-minute BTCUSDT USD-margined perpetual futures data from January 2022 to December 2025. Four forecasting models are compared: a persistence benchmark, AR(24), Fixed HAR-X, and TVP-HAR-X. The sample is divided chronologically into training, validation, and final test periods. Preprocessing parameters and model specifications are determined using the training and validation samples, and the 2025 test sample is reserved for a single final evaluation.

The study addresses two questions. First, do multi-horizon volatility components and high-frequency market activity improve one-hour-ahead forecasts relative to persistence and AR(24)? Second, does coefficient variation provide additional predictive value beyond a fixed-parameter HAR-X model using the same information set?

The results show that both HAR-X specifications substantially outperform persistence and AR(24). Fixed HAR-X achieves the lowest test QLIKE, while TVP-HAR-X records the lowest MAE and RMSE. Trading activity contributes the clearest incremental information among the additional predictors. Coefficient variation improves average forecast accuracy on the log scale, although it does not improve the primary variance-scale loss.

This study makes three contributions. It evaluates the proposed models against both persistence and a strong autoregressive benchmark; it combines multi-horizon realized volatility with hourly market-activity variables; and it compares fixed and time-varying coefficients using an identical predictor set. The last comparison isolates the forecasting value of parameter adaptation

from differences in observable information.

The remainder of the report is organized as follows. Section 2 describes the data and variable construction. Section 3 presents the forecasting design and evaluation criteria. Section 4 introduces the four models. Section 5 reports the empirical results, and Section 6 discusses their interpretation and limitations. Section 7 concludes.

2 Data and Variable Construction

2.1 Data Source and Sample Period

The analysis uses five-minute candlestick data for the BTCUSDT USD-margined perpetual futures contract obtained from the Binance Public Data repository (Binance, nd). The sample covers January 1, 2022 through December 31, 2025, with all timestamps recorded in Coordinated Universal Time (UTC).

The 48 monthly files contain 420,768 observations. Each record includes opening, high, low, and closing prices, trading volume, quote volume, the number of trades, and taker-buy volume, following the Binance Kline data structure (Binance, nd). The five-minute frequency provides sufficient intraday detail for constructing hourly volatility and activity measures while remaining computationally manageable.

Observations are sorted chronologically, checked for duplicate timestamps, and aligned to a regular five-minute calendar. The sample begins at 00:00 UTC on January 1, 2022 and ends at 23:55 UTC on December 31, 2025.

2.2 Five-Minute Returns

Let $P_{t,j}$ denote the closing price of the j-th five-minute interval within hour t. The corresponding log return is

$$r_{t,j} = \log P_{t,j} - \log P_{t,j-1}. \tag{1}$$

Each return is timestamped at the end of its five-minute interval and assigned to the hour in which the price change is completed. The sample contains 420,767 valid returns; the first observation is unavailable because no preceding closing price exists.

2.3 Hourly Realized Variance

Realized variance aggregates high-frequency squared returns to measure price variation over a specified interval (Andersen et al., 2003). The same approach has been applied to Bitcoin volatility forecasting using intraday data (Bergsli et al., 2022). Let $\mathcal{J}_t$ denote the twelve five-minute returns completed during hour t. Hourly realized variance is

$$RV_t = \sum_{j \in \mathcal{J}_t} r_{t,j}^2. \tag{2}$$

This measure incorporates the intrahour price path and is more informative about short-term variation than a single hourly return.

The aggregation produces 35,063 complete hourly bins. Because realized variance is positive and strongly right-skewed, the forecasting variable is defined as

$$y_t = \log(RV_t). \tag{3}$$

No additive constant is introduced before taking logarithms. Hours without a valid volatility measure are retained on the hourly calendar and marked as missing.

2.4 Data Quality Control

A zero-activity bar is defined as a five-minute interval with identical opening, high, low, and closing prices, zero trading volume, and zero recorded trades. The data contain 55 such bars, grouped into eight runs and affecting 11 hourly observations.

Affected hours remain on the original calendar, but their forecasting targets are set to missing. Lags and rolling variables are constructed before any model-ready filtering. This procedure preserves the temporal meaning of each lag and prevents observations separated by missing hours from being treated as adjacent.

After quality control, 35,052 hourly observations have valid realized variance and log realized variance.

2.5 Multi-Horizon Volatility Features

The HAR framework represents volatility persistence through aggregated components measured over different horizons (Corsi, 2009). This study adapts the framework to hourly forecasting using one-hour, six-hour, daily, and weekly components:

$$
\begin{aligned}
HAR_t^{(1)} &= y_t, \\
HAR_t^{(6)} &= \frac{1}{6}\sum_{i=0}^{5} y_{t-i}, \\
HAR_t^{(24)} &= \frac{1}{24}\sum_{i=0}^{23} y_{t-i}, \\
HAR_t^{(168)} &= \frac{1}{168}\sum_{i=0}^{167} y_{t-i}.
\end{aligned}
$$

These variables summarize current-hour, recent intraday, daily, and weekly volatility conditions. All rolling averages are calculated on the complete hourly calendar. A feature is marked as missing whenever its required window contains an invalid observation.

2.6 Downside Realized Semivariance

Realized semivariance separates total realized variation into components generated by negative and positive intraday returns (Barndorff-Nielsen et al., 2010). Hourly downside realized semivariance is defined as

$$
RS_t^- = \sum_{j \in \mathcal{J}_t} r_{t,j}^2 \mathbb{I}(r_{t,j} < 0), \tag{4}
$$

where $\mathbb{I}(\cdot)$ is the indicator function.

Because the raw measure is highly right-skewed, it is transformed using a training-based scale:

$$
DS_t = \log\left(1 + \frac{RS_t^-}{s_{\text{train}}}\right), \tag{5}
$$

where s_{train} is the median of positive downside-semivariance observations in the model-ready training sample. Its estimated value is $s_{\text{train}} = 5.2624359 \times 10^{-6}$. The same value is applied to the validation and test samples.

2.7 Trading Activity

Hourly trading activity is measured by the total number of trades across the twelve five-minute intervals within each hour. Because the distribution is strongly right-skewed, the transformed measure is $Trades_t = \log(1 + N_t)$, where N_t is the hourly trade count.

Trading activity provides a proxy for information arrival and market participation. The mixture-of-distributions framework links trading intensity and price variability through common information flows (Clark, 1973), while empirical evidence documents a close association between transaction counts and volatility (Jones et al., 1994). The variable is included to test whether current market activity predicts next-hour volatility after controlling for recent realized variance.

2.8 Aggressive-Buying Proxy

The Binance data report taker-buy base volume, representing volume initiated by market buy orders. The aggressive-buying proxy is

$$Buy_t = 2\left(\frac{TBV_t}{V_t}\right) - 1, \tag{6}$$

where TBV_t is hourly taker-buy base volume and V_t is total hourly base volume.

The variable is approximately bounded between -1 and 1. Positive values indicate that aggressive buying accounts for more than half of total volume, while negative values indicate stronger aggressive selling. Its sample mean is close to zero.

2.9 Feature Standardization

The HAR-X predictor set comprises

$$\left[HAR_t^{(1)},\, HAR_t^{(6)},\, HAR_t^{(24)},\, HAR_t^{(168)},\, DS_t,\, Trades_t,\, Buy_t\right].$$

Each predictor is standardized using the mean and standard deviation estimated from the model-ready training sample:

$$z_{k,t} = \frac{x_{k,t} - \bar{x}_{k,\text{train}}}{s_{k,\text{train}}}, \tag{7}$$

where $x_{k,t}$ is predictor k, $\bar{x}_{k,\text{train}}$ is its training-sample mean, and $s_{k,\text{train}}$ is its training-sample standard deviation.

The transformation parameters remain fixed during validation and test evaluation, preventing information from either period from entering the preprocessing stage.

2.10 Final Modeling Sample

The complete hourly calendar contains 35,063 rows. Requiring a valid one-hour-ahead target and complete values for all HAR-X features yields 33,705 observations for the common model comparison.

Table 1: Final model-ready sample by period

Period	Calendar observations	Model-ready observations
Training: 2022–2023	17,518	16,673
Validation: 2024	8,784	8,610
Test: 2025	8,760	8,422
Total used for modeling	35,062	33,705

The table excludes the final forecast origin because its one-hour-ahead target extends beyond the sample. That timestamp remains in the complete hourly calendar used for feature construction, which accounts for the difference between 35,063 calendar rows and the table total of 35,062.

3 Forecasting Design

3.1 One-Hour-Ahead Forecasting Problem

Let τ denote the forecast origin and $\mathcal{F}_\tau$ the information available at that time. The target is next-hour log realized variance, $y_{\tau+1} = \log(RV_{\tau+1})$. Each model produces the conditional forecast

$$\widehat{y}_{\tau+1|\tau} = E\left[y_{\tau+1} \mid \mathcal{F}_\tau\right]. \tag{8}$$

For a forecast issued at 12:00 UTC, the predictors use information observed by 12:00 UTC and the target is realized variance from 12:00 to 13:00 UTC.

Lagged variables and rolling averages are constructed on the complete hourly calendar before model-ready observations are selected. This preserves the calendar interpretation of each lag and rolling window when invalid hours occur.

3.2 Chronological Sample Division

The sample is divided chronologically into training, validation, and test periods.

Table 2: Chronological forecasting design

Sample	Period	Purpose
Training	January 2022–December 2023	Estimation and preprocessing
Validation	January 2024–December 2024	Model and hyperparameter selection
Test	January 2025–December 2025	Final out-of-sample evaluation

The training sample determines preprocessing parameters and initial model estimates. The downside-semivariance scale and feature-standardization parameters are estimated exclusively from this period.

The validation sample is used to select the autoregressive order and evaluate the fixed and time-varying specifications. After the model structures are fixed, the selected models are re-estimated using the combined training and validation sample. The 2025 observations are reserved for the final evaluation.

3.3 Common Evaluation Sample

All models are compared on identical forecast timestamps. An observation is retained only when the one-hour-ahead target and all predictors required by HAR-X are available.

The common sample contains 16,673 training observations, 8,610 validation observations, and 8,422 test observations.

Persistence and AR(24) could generate forecasts for additional timestamps. Restricting evaluation to the common sample ensures that differences in reported losses reflect forecast performance under the same market conditions.

3.4 Sequential Forecasting and Information Control

The fixed-parameter models use predictor values observed at the forecast origin and retain constant coefficient estimates throughout each evaluation period.

TVP-HAR-X generates forecasts sequentially. At origin τ, the predicted state produces $\widehat{y}_{\tau+1|\tau}$. Once $y_{\tau+1}$ becomes available, the Kalman filter updates the state distribution for the following forecast.

Invalid target hours remain on the state-space calendar as missing observations. The state transition continues through these hours, while the measurement update is omitted.

3.5 Model Selection Rules

The autoregressive order is selected from $p \in \{1, 2, \ldots, 24\}$. Each candidate model is estimated on the training sample and evaluated on the same validation timestamps.

Validation QLIKE is the primary selection criterion. Ties are resolved sequentially using:

(i) lower validation RMSE;

(ii) lower validation MAE;

(iii) lower autoregressive order.

Training-sample AIC and BIC are reported as supplementary diagnostics. The HAR-X predictor set and TVP-HAR-X state specification are fixed before the test evaluation.

3.6 Mean Absolute Error

Mean absolute error is calculated on the log realized-variance scale:

$$MAE = \frac{1}{N} \sum_{t=1}^{N} |y_t - \widehat{y}_t| \,. \tag{9}$$

MAE reports the average absolute forecast error and gives each observation a linear contribution to aggregate loss.

3.7 Root Mean Squared Error

Root mean squared error is

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^{N} (y_t - \widehat{y}_t)^2}. \tag{10}$$

The squared-error transformation assigns greater weight to large forecast errors.

3.8 QLIKE Loss

QLIKE is the primary volatility forecast loss. It is suitable for comparing forecasts when realized variance is an imperfect proxy for latent conditional variance (Patton, 2011).

Let $RV_t = \exp(y_t)$ and $\widehat{RV}_t = \exp(\widehat{y}_t)$. Observation-level QLIKE is

$$QLIKE_t = \frac{RV_t}{\widehat{RV}_t} - \log\left(\frac{RV_t}{\widehat{RV}_t}\right) - 1. \tag{11}$$

On the log scale, the same loss can be written as

$$QLIKE_t = \exp\left(y_t - \widehat{y}_t\right) - \left(y_t - \widehat{y}_t\right) - 1. \tag{12}$$

Average QLIKE is

$$QLIKE = \frac{1}{N}\sum_{t=1}^{N} QLIKE_t. \tag{13}$$

Lower values indicate better performance. QLIKE evaluates proportional variance errors and can impose a large penalty when high realized variance is substantially underestimated.

3.9 Forecast Loss Comparison

The percentage improvement of model m over benchmark b is

$$Improvement_{m,b} = 100\left(\frac{L_b - L_m}{L_b}\right), \tag{14}$$

where L_b and L_m denote average forecast losses. A positive value indicates lower loss for model m.

Predictive accuracy is assessed through the loss-differential framework of Diebold and Mariano (1995). For TVP-HAR-X and a benchmark model b, the loss differential is

$$d_t = L_{b,t} - L_{\mathrm{TVP},t}. \tag{15}$$

Positive values favor TVP-HAR-X. The hypotheses are

$$H_0 : E(d_t) \leq 0, \qquad H_1 : E(d_t) > 0.$$

Because forecast losses may be heteroskedastic and serially correlated, the mean loss differential is evaluated using Newey–West HAC standard errors with 24 lags (Newey and West, 1987). The resulting statistic is interpreted as a Diebold–Mariano-type test of equal predictive accuracy.

The pre-specified primary comparison evaluates TVP-HAR-X against Fixed HAR-X under QLIKE. Absolute-error and squared-error comparisons are reported as supplementary evidence.

4 Methodology

4.1 Modeling Strategy

The empirical analysis compares four models of increasing flexibility. Each specification adds a distinct source of predictive information or parameter variation:

$$\text{Persistence} \longrightarrow \text{AR} \longrightarrow \text{Fixed HAR-X} \longrightarrow \text{TVP-HAR-X}.$$

The persistence benchmark uses only current-hour volatility. The autoregressive model estimates a flexible pattern of short-run lag dependence. Fixed HAR-X replaces individual lags with multi-horizon volatility summaries and supplements them with market-activity variables. TVP-HAR-X retains the same predictors while allowing their coefficients to evolve over time.

This sequence separates three potential sources of forecast improvement:

(i) persistence beyond the current hour;

(ii) multi-horizon volatility and market-activity information;

(iii) time variation in predictor coefficients.

Table 3 summarizes the role of each model.

Table 3: Forecasting models and their empirical roles

Model	Main information	Empirical question
Persistence	Current-hour volatility	Does the current volatility level provide an adequate one-hour-ahead forecast?
AR(24)	Twenty-four hourly lags	Does a flexible short-run lag structure improve forecast accuracy?
Fixed HAR-X	Multi-horizon volatility and market-activity features	Do structured long-horizon summaries and current market conditions add predictive information?
TVP-HAR-X	The HAR-X predictor set with evolving coefficients	Does gradual coefficient variation improve forecasts relative to a fixed relationship?

4.2 Persistence Benchmark

The persistence forecast is

$$\widehat{y}_{t+1|t}^{\text{Naive}} = y_t. \tag{16}$$

On the original variance scale, this implies $\widehat{RV}_{t+1|t}^{\text{Naive}} = RV_t$.

Volatility persistence makes this a relevant benchmark for short-horizon forecasting. The model contains no estimated parameters and represents the performance available from the most recent observed volatility alone.

4.3 Fixed-Parameter Autoregressive Model

The general one-hour-ahead AR(p) model is

$$y_{t+1} = c + \sum_{j=1}^{p} \phi_j y_{t-j+1} + \varepsilon_{t+1}, \tag{17}$$

where c is an intercept, ϕ_j is the coefficient on lag j, and ε_{t+1} is a zero-mean forecast disturbance.

Under this indexing convention,

$$y_{t-j+1} = \begin{cases} y_t, & j = 1, \\ y_{t-1}, & j = 2, \\ \vdots & \\ y_{t-p+1}, & j = p. \end{cases}$$

The corresponding forecast is

$$\widehat{y}_{t+1|t}^{\text{AR}(p)} = \widehat{c} + \sum_{j=1}^{p} \widehat{\phi}_j y_{t-j+1}. \tag{18}$$

The persistence benchmark is obtained from the restrictions $c = 0$ and $\phi_1 = 1$. Estimating the coefficients allows the model to capture mean reversion and distribute predictive weight across several recent hours.

4.3.1 Stationarity Assessment

The AR analysis begins with an assessment of the training-period log realized-variance series. The graphical diagnostics include the series itself, a 168-hour rolling mean, a 168-hour rolling standard deviation, the autocorrelation function, and the partial autocorrelation function.

The augmented Dickey–Fuller test accommodates higher-order serial dependence through lagged differences (Said and Dickey, 1984). Its null hypothesis is

$$H_0^{\mathrm{ADF}} : \text{the series contains a unit root.}$$

The KPSS test uses level stationarity as the null hypothesis (Kwiatkowski et al., 1992):

$$H_0^{\mathrm{KPSS}} : \text{the series is level stationary.}$$

The two tests provide complementary evidence. Joint rejection can arise when a series is mean reverting but exhibits changes in its level or dependence structure over long periods.

Formal tests are applied to the longest continuous valid training segment. This avoids closing gaps by treating nonadjacent observations as consecutive.

4.3.2 Order Selection and Estimation

Candidate orders are restricted to $p \in \{1, 2, \ldots, 24\}$. The upper bound covers one complete day of hourly dependence while limiting the number of estimated parameters.

Each candidate model is estimated by ordinary least squares on the training sample. AIC and BIC are reported as in-sample diagnostics, while validation QLIKE determines the selected order. Validation RMSE and MAE serve as secondary criteria.

The procedure selects $p = 24$, yielding

$$y_{t+1} = c + \sum_{j=1}^{24} \phi_j y_{t-j+1} + \varepsilon_{t+1}. \tag{19}$$

Dynamic stability is assessed through the roots of the characteristic polynomial

$$1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_{24} z^{24} = 0. \tag{20}$$

The model is stable when all roots lie outside the unit circle. Residual dependence is evaluated using autocorrelations and Ljung–Box statistics at several horizons.

AR(24) serves as the principal fixed-lag benchmark for the subsequent HAR-X comparisons.

4.4 The HAR-X Specification

AR(24) estimates a separate coefficient for each of the previous 24 hourly observations and contains no direct summary of volatility beyond one day. The HAR framework represents persistent volatility through aggregated components measured over several horizons (Corsi, 2009).

The volatility predictor vector is

$$\mathbf{h}_t = \begin{bmatrix} HAR_t^{(1)} \\ HAR_t^{(6)} \\ HAR_t^{(24)} \\ HAR_t^{(168)} \end{bmatrix}.$$

The one-hour and six-hour components describe recent conditions, while the 24-hour and 168-hour components summarize daily and weekly volatility. This construction compresses the lag history into four interpretable variables and avoids estimating a separate coefficient for every hourly lag.

Three market-activity predictors are added:

$$\mathbf{m}_t = \begin{bmatrix} DS_t \\ Trades_t \\ Buy_t \end{bmatrix}.$$

Trading activity is motivated by the relationship between information arrival, transaction intensity, and price variability (Clark, 1973; Jones et al., 1994). Downside realized semivariance measures variation generated by negative intraday returns (Barndorff-Nielsen et al., 2010). The aggressive-buying proxy summarizes the direction of taker-initiated volume.

These variables are derived from aggregated candlestick data. They represent market-activity and order-flow proxies rather than direct measures of bid–ask spreads, market depth, or other limit-order-book conditions.

4.5 Fixed HAR-X Model

Let the standardized predictor vector be

$$\mathbf{x}_t = \begin{bmatrix} 1 \\ z_{HAR^{(1)},t} \\ z_{HAR^{(6)},t} \\ z_{HAR^{(24)},t} \\ z_{HAR^{(168)},t} \\ z_{DS,t} \\ z_{Trades,t} \\ z_{Buy,t} \end{bmatrix}. \tag{21}$$

The first element is an intercept, and the remaining variables are standardized using training-sample moments.

The fixed-parameter HAR-X model is

$$y_{t+1} = \mathbf{x}_t^{\mathsf{T}}\boldsymbol{\beta} + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, H), \tag{22}$$

with coefficient vector

$$\boldsymbol{\beta} = \begin{bmatrix} \mu & \beta_1 & \beta_6 & \beta_{24} & \beta_{168} & \beta_{DS} & \beta_{Trades} & \beta_{Buy} \end{bmatrix}^{\mathsf{T}}.$$

Equivalently,

$$\begin{aligned} y_{t+1} = \mu &+ \beta_1 z_{HAR^{(1)},t} + \beta_6 z_{HAR^{(6)},t} + \beta_{24} z_{HAR^{(24)},t} \\ &+ \beta_{168} z_{HAR^{(168)},t} + \beta_{DS} z_{DS,t} + \beta_{Trades} z_{Trades,t} + \beta_{Buy} z_{Buy,t} + \varepsilon_{t+1}. \end{aligned} \tag{23}$$

The coefficients are estimated by ordinary least squares. Their interpretation is conditional on the remaining predictors. The HAR terms measure incremental dependence at different volatility horizons, while the three additional variables capture information associated with the composition and intensity of current market activity.

Coefficient inference uses Newey–West heteroskedasticity- and autocorrelation-consistent standard errors with 24 lags (Newey and West, 1987). These standard errors affect inference but leave the OLS coefficient estimates and forecasts unchanged.

Fixed HAR-X provides the benchmark for evaluating the value of time-varying coefficients under an identical observable information set.

4.6 Motivation for Time-Varying Parameters

Fixed HAR-X imposes a constant coefficient vector, $\boldsymbol{\beta}_t = \boldsymbol{\beta}$, throughout the sample. This restriction may be strong for Bitcoin markets, where volatility conditions, participation, and trading activity can change substantially. Previous evidence also shows that jumps and structural breaks affect Bitcoin volatility forecasts (Shen et al., 2020).

A constant coefficient summarizes the average predictive relationship over the estimation period. When that relationship changes, a fixed model may adjust slowly and combine observations generated under different market conditions.

TVP-HAR-X treats the coefficient vector as a latent dynamic state. Coefficients evolve gradually as new forecast errors alter the estimated state distribution. This structure permits adaptation while preserving continuity between adjacent forecast origins.

4.7 TVP-HAR-X State-Space Model

TVP-HAR-X uses the same observable predictors as Fixed HAR-X and allows their coefficients to vary over time. The model is written in state-space form, linking the observed target to a latent coefficient vector (Durbin and Koopman, 2012).

The observation equation is

$$y_{t+1} = \mathbf{x}_t^{\mathsf{T}} \boldsymbol{\beta}_t + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, H). \tag{24}$$

The state vector is

$$\boldsymbol{\beta}_t = \begin{bmatrix} \mu_t \\ \beta_{1,t} \\ \beta_{6,t} \\ \beta_{24,t} \\ \beta_{168,t} \\ \beta_{DS,t} \\ \beta_{Trades,t} \\ \beta_{Buy,t} \end{bmatrix}. \tag{25}$$

Its transition equation is a random walk:

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \boldsymbol{\eta}_{t+1}, \qquad \boldsymbol{\eta}_{t+1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}). \tag{26}$$

The state covariance matrix is restricted to

$$\mathbf{Q} = \operatorname{diag}(q_\mu, q_\beta, q_\beta, q_\beta, q_\beta, q_\beta, q_\beta, q_\beta). \tag{27}$$

The intercept has innovation variance q_μ, and the seven slope coefficients share q_β. This restriction reduces the variance parameters to H, q_μ, and q_β, providing a parsimonious specification of coefficient variation.

The state variances govern the rate of adaptation. Values close to zero produce slowly changing coefficients, while larger values permit faster responses to new observations. All three variance parameters are estimated from the data.

4.8 Kalman Filtering and Sequential Forecasting

Conditional on the variance parameters, TVP-HAR-X is a linear Gaussian state-space model. The Kalman filter recursively produces forecasts and updates the conditional distribution of the latent coefficients (Kalman, 1960; Durbin and Koopman, 2012).

Let

$$\mathbf{a}_{t|t-1} = E\left[\boldsymbol{\beta}_t \mid \mathcal{F}_{t-1}\right] \tag{28}$$

denote the predicted state, with covariance

$$\mathbf{P}_{t|t-1} = \mathrm{Var}\left(\boldsymbol{\beta}_t \mid \mathcal{F}_{t-1}\right). \tag{29}$$

Under the random-walk transition,

$$\mathbf{a}_{t|t-1} = \mathbf{a}_{t-1|t-1}, \tag{30}$$

and

$$\mathbf{P}_{t|t-1} = \mathbf{P}_{t-1|t-1} + \mathbf{Q}. \tag{31}$$

The one-hour-ahead forecast is

$$\widehat{y}_{t+1|t} = \mathbf{x}_t^{\mathsf{T}} \mathbf{a}_{t|t-1}, \tag{32}$$

with conditional forecast variance

$$F_t = \mathbf{x}_t^{\mathsf{T}} \mathbf{P}_{t|t-1} \mathbf{x}_t + H. \tag{33}$$

After y_{t+1} becomes available, the forecast innovation is

$$v_t = y_{t+1} - \widehat{y}_{t+1|t}, \tag{34}$$

and the Kalman gain is

$$\mathbf{K}_t = \frac{\mathbf{P}_{t|t-1}\mathbf{x}_t}{F_t}. \tag{35}$$

The filtered state and covariance are updated according to

$$\mathbf{a}_{t|t} = \mathbf{a}_{t|t-1} + \mathbf{K}_t v_t, \tag{36}$$

and

$$\mathbf{P}_{t|t} = \mathbf{P}_{t|t-1} - \mathbf{K}_t \mathbf{x}_t^{\mathsf{T}} \mathbf{P}_{t|t-1}. \tag{37}$$

The magnitude of each update depends on the forecast innovation, state uncertainty, predictor values, and observation variance. The filter therefore regulates coefficient adjustment through the Kalman gain.

4.9 Missing Observations

Invalid target hours remain on the hourly calendar. During such an hour, the state prediction proceeds according to

$$\mathbf{a}_{t|t-1} = \mathbf{a}_{t-1|t-1}, \qquad \mathbf{P}_{t|t-1} = \mathbf{P}_{t-1|t-1} + \mathbf{Q}.$$

Because no valid target is observed, the measurement update is omitted:

$$\mathbf{a}_{t|t} = \mathbf{a}_{t|t-1}, \qquad \mathbf{P}_{t|t} = \mathbf{P}_{t|t-1}.$$

This treatment preserves elapsed calendar time across missing hours.

4.10 Parameter Estimation

The initial state mean is set to the Fixed HAR-X coefficient estimate. The initial covariance is

$$\mathbf{P}_0 = 0.25\mathbf{I}_8, \tag{38}$$

where $\mathbf{I}_8$ is the 8×8 identity matrix.

The variance-parameter vector is $\boldsymbol{\theta} = (H, q_\mu, q_\beta)^\mathsf{T}$. It is estimated by maximizing the Kalman-filter prediction-error likelihood (Durbin and Koopman, 2012).

Conditional on information available before observation t,

$$v_t \sim \mathcal{N}(0, F_t). \tag{39}$$

Ignoring constants, the log likelihood is

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2} \sum_{t \in \mathcal{T}_{\text{valid}}} \left[\log(F_t) + \frac{v_t^2}{F_t} \right], \tag{40}$$

where $\mathcal{T}_{\text{valid}}$ contains observations with valid targets.

Positivity is imposed through exponential transformations of the variance parameters. Optimization is performed from several initial values, and the converged solution with the highest likelihood is retained.

During model development, the variance parameters are estimated on the training sample and evaluated sequentially on validation data. After the specification is fixed, they are re-estimated using the combined training and validation period for the final test forecasts.

4.11 Interpretation of the Time-Varying Coefficients

The estimated state paths describe conditional predictive associations. Causal interpretation lies outside the scope of the model.

For example, a positive value of $\beta_{Trades,t}$ indicates that higher trading activity is associated with higher next-hour log realized variance at time t, conditional on the remaining predictors. Changes in $\beta_{1,t}$ and $\beta_{168,t}$ describe variation in the relative predictive importance of recent and weekly volatility.

The state estimates are used to assess coefficient stability, identify predictors with substantial temporal variation, and compare their behavior across market conditions.

4.12 Model Diagnostics

AR(24) diagnostics include characteristic roots, residual autocorrelations, and Ljung–Box portmanteau statistics (Ljung and Box, 1978).

For Fixed HAR-X, the analysis reports the residual mean, variance, first-order autocorrelation, Durbin–Watson statistic, and Ljung–Box statistics.

TVP-HAR-X diagnostics use standardized one-step-ahead innovations,

$$u_t = \frac{v_t}{\sqrt{F_t}}. \tag{41}$$

Under the linear Gaussian specification, these innovations should have approximately zero mean, unit variance, and limited serial correlation.

Because the diagnostic samples are large, small residual correlations may produce statistically significant portmanteau tests. The test results are interpreted jointly with residual magnitudes, short-lag autocorrelations, and out-of-sample forecast performance.

4.13 Final Model Comparison

Model selection is conducted using the training and validation periods. The selected AR(24), Fixed HAR-X, and TVP-HAR-X specifications are then re-estimated on the combined development sample and evaluated on the same 2025 test timestamps.

The comparison addresses three questions:

(1) Does AR(24) improve on volatility persistence?

(2) Does Fixed HAR-X improve on AR(24) through multi-horizon and market-activity information?

(3) Does coefficient variation improve on Fixed HAR-X under an identical predictor set?

The final comparison isolates the contribution of parameter variation because Fixed HAR-X and TVP-HAR-X use the same observable information.

5 Empirical Results

5.1 Stationarity and Dependence in Log Realized Variance

Stationarity diagnostics are conducted using the training sample. Figure 1 presents hourly log realized variance together with its 168-hour rolling mean.

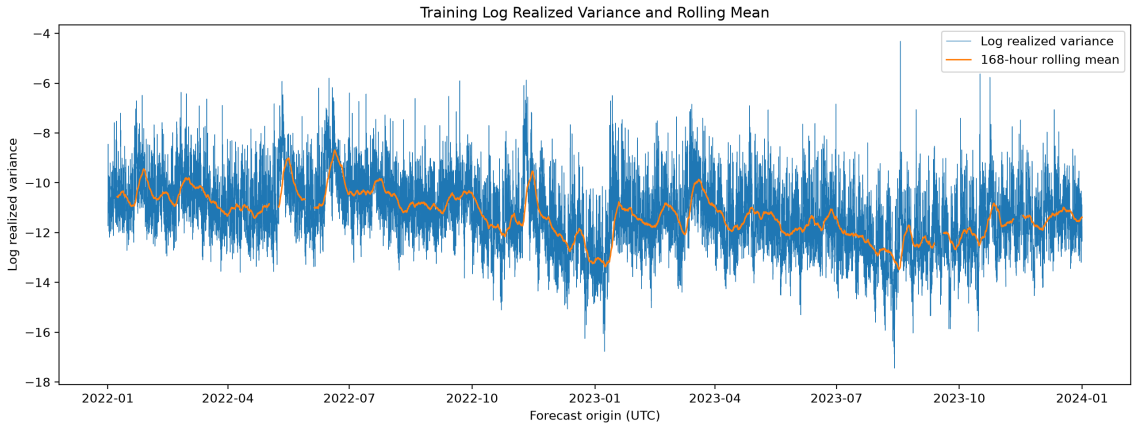


Figure 1: Training-period log realized variance and its 168-hour rolling mean

Log realized variance repeatedly returns toward its prevailing level after large increases, indicating mean-reverting behavior. The rolling mean nevertheless changes over the training period, suggesting that the unconditional level is not constant throughout the sample.

Table 4 reports the formal test results.

Table 4: Training-period stationarity tests

Test	Statistic	p-value	Lags	Observations	Decision at 5%
ADF	-8.0186	< 0.001	40	11,277	Reject unit root
KPSS	5.9013	0.010	57	11,318	Reject level stationarity

The augmented Dickey–Fuller test strongly rejects the unit-root null (Said and Dickey, 1984), providing evidence of mean reversion. The KPSS test also rejects global level stationarity

(Kwiatkowski et al., 1992). Taken together, the results indicate a persistent and mean-reverting series whose level or dependence structure changes over longer periods.

These findings support the use of a fixed autoregression as a forecasting benchmark while also motivating a specification that allows gradual parameter variation.

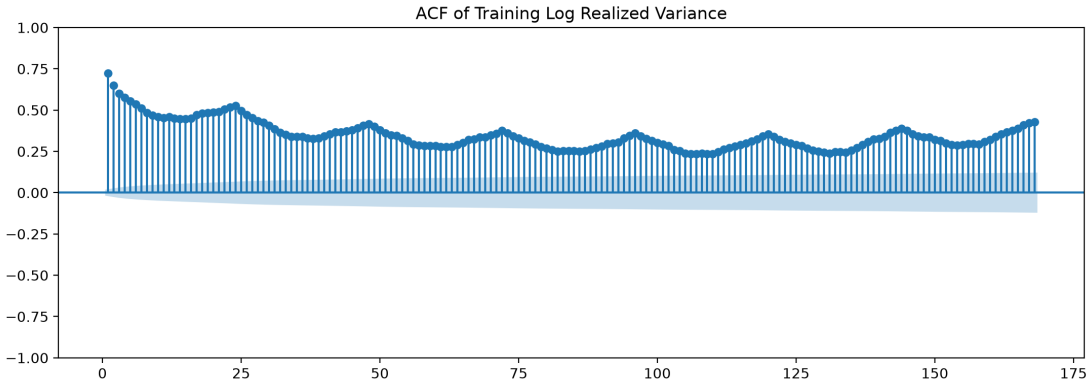


Figure 2: Autocorrelation function of training-period log realized variance

Figure 2 shows substantial autocorrelation beyond the first few hours. The persistence benchmark therefore captures only part of the temporal dependence in hourly volatility.

5.2 Autoregressive Order Selection

AR orders from 1 through 24 are estimated on the training sample and ranked by validation QLIKE. Table 5 reports the five leading specifications.

Table 5: Leading AR specifications ranked by validation QLIKE

Order	Training AIC	Training BIC	Validation MAE	Validation RMSE	Validation QLIKE
24	43,353.53	43,546.57	0.6219	0.8023	0.5105
23	43,392.98	43,578.29	0.6232	0.8037	0.5122
22	43,455.91	43,633.50	0.6253	0.8061	0.5143
21	43,528.03	43,697.90	0.6267	0.8080	0.5161
20	43,557.89	43,720.04	0.6277	0.8090	0.5196

AR(24) achieves the lowest validation QLIKE, RMSE, and MAE and is selected as the autoregressive benchmark. Its position at the upper boundary of the candidate set suggests that information from the preceding 24 hours remains useful. The incremental gains decline as the order increases, indicating that predictive dependence is distributed across several lags rather than concentrated in a single hour.

The validation losses for AR(24) are 0.6219 for MAE, 0.8023 for RMSE, and 0.5105 for QLIKE. Relative to persistence, these values represent reductions of approximately 13.14%, 12.59%, and 21.22%, respectively. Estimated lag dynamics therefore provide substantial predictive value beyond the current volatility level.

5.2.1 AR Stability and Residual Diagnostics

All characteristic roots of AR(24) lie outside the unit circle. The minimum root modulus is 1.0084, indicating a stable but highly persistent specification.

The first-order residual autocorrelation is 0.0036. Ljung–Box portmanteau tests nevertheless reject residual white noise at lags 29, 48, and 168, with p-values below 0.001 (Ljung and Box, 1978).

AR(24) captures most immediate dependence but leaves weaker correlation at longer horizons. It therefore provides a strong forecasting benchmark while leaving scope for the multi-horizon HAR-X specification.

5.3 Fixed HAR-X Estimation Results

Fixed HAR-X combines multi-horizon volatility components with downside semivariance, trading activity, and aggressive-buying information. Table 6 reports standardized coefficients with Newey–West HAC standard errors (Newey and West, 1987).

Table 6: Fixed HAR-X standardized coefficients with HAC(24) inference

Predictor	Coefficient	HAC standard error	t-statistic	p-value
Current-hour volatility	0.2616	0.0239	10.94	< 0.001
Six-hour volatility	0.2632	0.0167	15.73	< 0.001
Twenty-four-hour volatility	0.2574	0.0183	14.09	< 0.001
One-hundred-sixty-eight-hour volatility	0.1163	0.0140	8.31	< 0.001
Transformed downside semivariance	-0.0062	0.0192	-0.32	0.745
Log number of trades	0.3379	0.0153	22.13	< 0.001
Aggressive-buying proxy	-0.0006	0.0088	-0.07	0.948

All four HAR components have positive and statistically significant coefficients, consistent with the heterogeneous-horizon representation of realized volatility (Corsi, 2009). Current, intraday, daily, and weekly volatility each contribute information after conditioning on the remaining predictors.

Trading activity has a coefficient of 0.3379 and is highly significant. Because the predictors are standardized, a one-standard-deviation increase in log trade count is associated with an increase of approximately 0.338 in next-hour log realized variance, conditional on the other variables.

This result is consistent with the information-arrival interpretation of trading activity (Clark, 1973) and the documented relationship between transaction counts and volatility (Jones et al., 1994). Among the three additional market-activity variables, trade count provides the strongest incremental predictive evidence.

The coefficients on downside semivariance and aggressive buying are small and statistically insignificant. Their average conditional associations with next-hour volatility are therefore weak over the estimation period.

All coefficient estimates are interpreted as predictive associations.

5.4 Fixed HAR-X Validation Performance

Table 7 compares the four models on the validation sample.

Table 7: Validation-period forecast performance

Model	MAE	RMSE	QLIKE	QLIKE improvement vs. Naive
Naive	0.7160	0.9179	0.6480	0.00%
AR(24)	0.6219	0.8023	0.5105	21.22%
Fixed HAR-X	0.6106	0.7913	0.5083	21.56%
TVP-HAR-X	0.6079	0.7857	0.4864	24.93%

Fixed HAR-X improves on AR(24) under all three metrics. The QLIKE reduction is 0.42%, showing that AR(24) already captures much of the predictive content in recent volatility.

Fixed HAR-X achieves this performance with seven structured predictors and incorporates information extending to the preceding week. The specification therefore provides a parsimonious and interpretable alternative to a larger set of individual hourly lags.

Its residual first-order autocorrelation is close to zero, and the Durbin–Watson statistic is 2.009. Ljung–Box tests remain significant at lags 24, 48, and 168, indicating residual dependence at longer horizons.

5.5 TVP-HAR-X Parameter Estimates

The TVP-HAR-X variance parameters are estimated on the training sample by maximizing the Kalman-filter prediction-error likelihood (Durbin and Koopman, 2012). Three sets of starting values converge to nearly identical solutions, indicating numerical stability.

Table 8: Training-period TVP-HAR-X variance parameters

Parameter	Variance	Standard deviation
Observation disturbance	7.4918×10^{-1}	8.6555×10^{-1}
Intercept-state innovation	2.5406×10^{-5}	5.0404×10^{-3}
Common slope-state innovation	1.6681×10^{-6}	1.2915×10^{-3}

Both state innovation variances are small relative to the observation variance, producing gradual coefficient movement. The intercept innovation standard deviation exceeds the common slope innovation standard deviation, indicating faster variation in the overall volatility level than in the marginal predictor effects.

The estimated state variances are positive, so the fitted model does not collapse numerically to a fixed-coefficient specification.

5.6 Time-Varying Coefficient Behavior

Table 9 summarizes the predicted state estimates during validation.

Table 9: Validation-period time-varying coefficient summary

State	Mean	Standard deviation	Minimum	Maximum
Intercept	-11.2035	0.0936	-11.4340	-10.9837
Current-hour volatility	0.1102	0.0312	0.0502	0.1918
Six-hour volatility	0.2447	0.0194	0.1988	0.2976
Twenty-four-hour volatility	0.1721	0.0271	0.1203	0.2514
One-hundred-sixty-eight-hour volatility	0.0076	0.0316	-0.0781	0.0576
Transformed downside semivariance	-0.0175	0.0508	-0.1261	0.0787
Log number of trades	0.5336	0.0528	0.4407	0.6513
Aggressive-buying proxy	-0.0324	0.0232	-0.0864	0.0373

Trading activity has the largest average positive coefficient and remains positive throughout validation, ranging from approximately 0.441 to 0.651. It is therefore a consistently informative predictor within this period.

The six-hour and twenty-four-hour coefficients also remain positive and vary within relatively narrow ranges. Short- and medium-horizon volatility persistence is consequently stable during validation.

The weekly coefficient fluctuates around zero, while the downside-semivariance and aggressive-buying coefficients change sign. Their predictive contributions vary more substantially across the sample.

The state paths describe changes in conditional forecasting relationships and are not assigned a causal interpretation.

5.7 TVP-HAR-X Validation Performance

TVP-HAR-X records validation MAE of 0.6079, RMSE of 0.7857, and QLIKE of 0.4864. QLIKE is 4.30% lower than for Fixed HAR-X and 24.93% lower than for the persistence benchmark.

The standardized innovations have a mean of approximately -0.0095, a standard deviation of 1.024, and first-order autocorrelation of 0.0007. These values are close to the corresponding theoretical benchmarks of zero, one, and zero.

Longer-horizon Ljung–Box tests remain significant. Given the large diagnostic sample, these rejections reflect small autocorrelations accumulated across several lags. The state-space model improves validation forecasts while leaving some temporal dependence unexplained.

5.8 Final Out-of-Sample Test Results

After model selection, AR(24), Fixed HAR-X, and TVP-HAR-X are re-estimated using the combined training and validation sample. The locked specifications are then evaluated on 8,422 common test observations from 2025.

Table 10 reports the final out-of-sample performance.

Table 10: Final 2025 test-set forecast performance

Model	MAE	MAE rank	RMSE	RMSE rank	QLIKE	QLIKE rank
Naive	0.7021	4	0.9101	4	0.7007	4
AR(24)	0.6133	3	0.7997	3	0.5054	3
Fixed HAR-X	0.5980	2	0.7807	2	**0.4764**	**1**
TVP-HAR-X	**0.5901**	**1**	**0.7734**	**1**	0.4800	2

All three estimated models improve substantially on persistence. Relative to Naive, test QLIKE falls by 27.87% for AR(24), 32.02% for Fixed HAR-X, and 31.50% for TVP-HAR-X.

Fixed HAR-X reduces QLIKE by 5.75% relative to AR(24). The structured multi-horizon and market-activity predictors therefore provide additional forecasting information beyond the 24 hourly lags.

The ranking of the two HAR-X models varies across loss functions. TVP-HAR-X records the lowest MAE and RMSE, while Fixed HAR-X records the lowest QLIKE. Relative to Fixed HAR-X, TVP-HAR-X reduces absolute error by 1.33% and squared error by 1.87%. Its QLIKE is 0.76% higher.

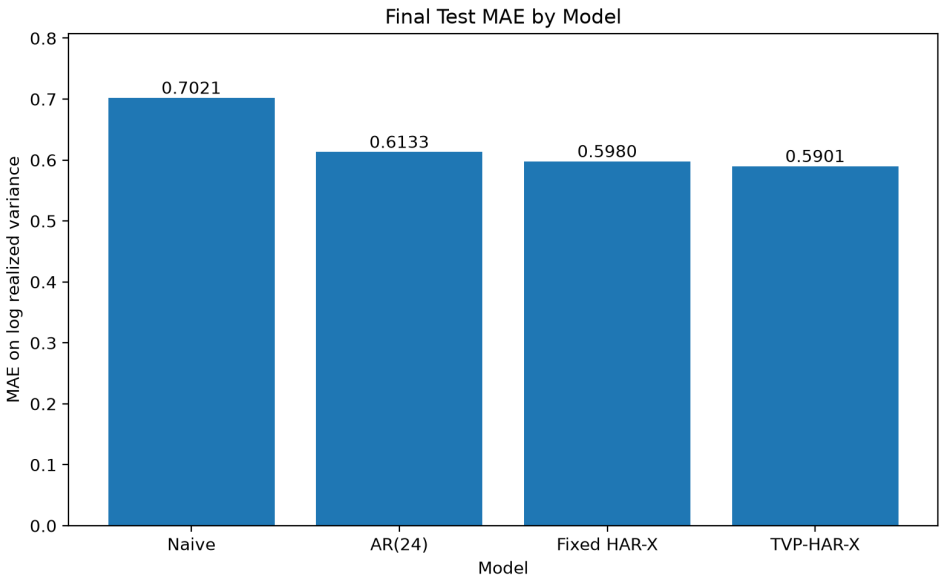


Figure 3: Final test MAE across forecasting models

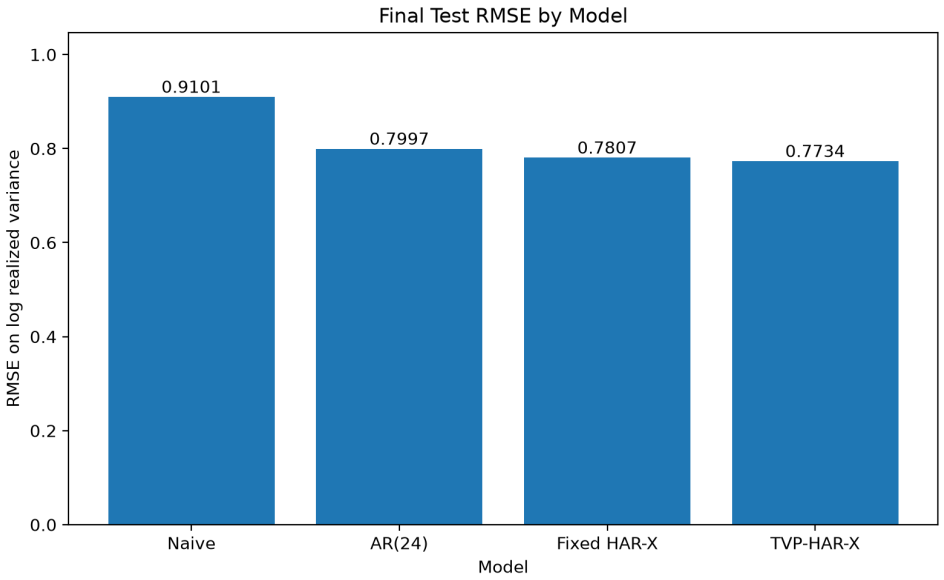


Figure 4: Final test RMSE across forecasting models

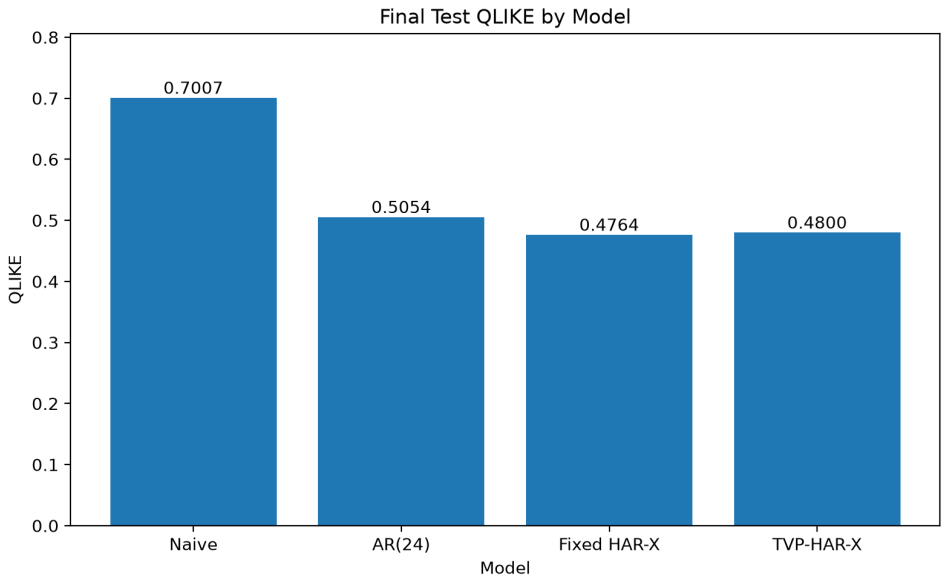


Figure 5: Final test QLIKE across forecasting models

Figures 3–5 show that the difference between the two HAR-X specifications is small relative to their gains over Naive and AR(24). They also illustrate the change in ranking between log-scale errors and variance-scale loss.

5.9 Predictive-Loss Significance Tests

Table 11 reports Diebold–Mariano-type loss-differential tests (Diebold and Mariano, 1995) using Newey–West HAC standard errors (Newey and West, 1987). The loss differential is defined so that positive values favor TVP-HAR-X.

Table 11: HAC loss-differential tests: TVP-HAR-X versus Fixed HAR-X

Loss	Fixed HAR-X	TVP-HAR-X	Improvement	Test statistic	One-sided p-value
Absolute error	0.5980	0.5901	1.33%	6.2596	< 0.001
Squared error	0.6096	0.5981	1.87%	6.0210	< 0.001
QLIKE	0.4764	0.4800	-0.76%	-1.2118	0.887

TVP-HAR-X significantly reduces absolute and squared loss relative to Fixed HAR-X. The corresponding one-sided p-values are below 0.001.

The QLIKE loss differential has a mean of -0.00362 and a one-sided p-value of 0.887. The test therefore provides no evidence that TVP-HAR-X improves on Fixed HAR-X under the pre-specified primary loss. The negative point estimate is consistent with the lower QLIKE reported for Fixed HAR-X.

TVP-HAR-X also improves significantly on Naive and AR(24) under all three losses. Its test QLIKE is approximately 5.03% lower than that of AR(24), with a one-sided p-value below 0.001.

5.10 Validation–Test Comparison

The relative performance of Fixed HAR-X and TVP-HAR-X changes between the validation and test periods. TVP-HAR-X reduces validation QLIKE by 4.30% relative to Fixed HAR-X, whereas its test QLIKE is 0.76% higher.

This change in ranking shows that the validation advantage of the time-varying specification does not persist uniformly through the test period.

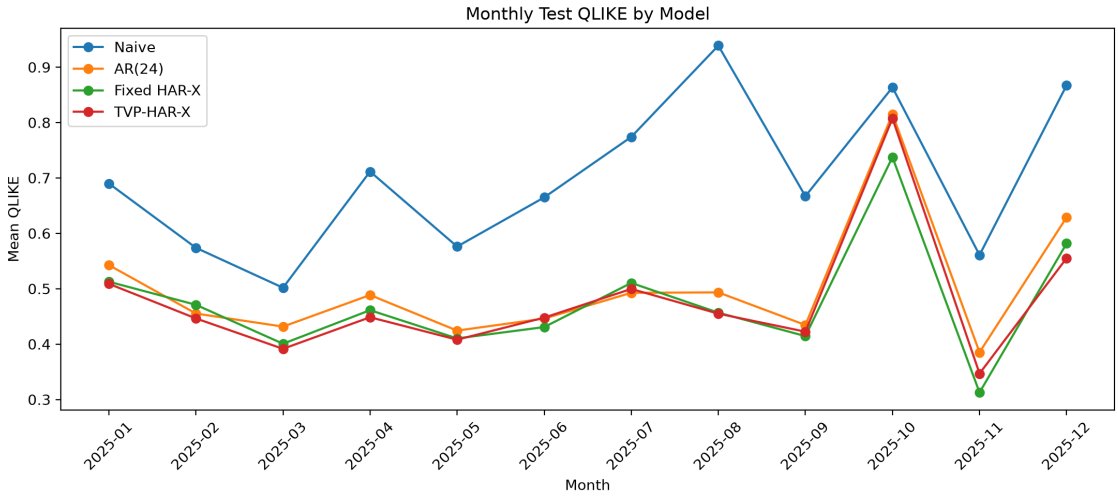


Figure 6: Monthly test-period QLIKE by model

Figure 6 reports monthly QLIKE during 2025. The ranking of Fixed HAR-X and TVP-HAR-X varies across months, so the annual averages combine periods with different relative performance.

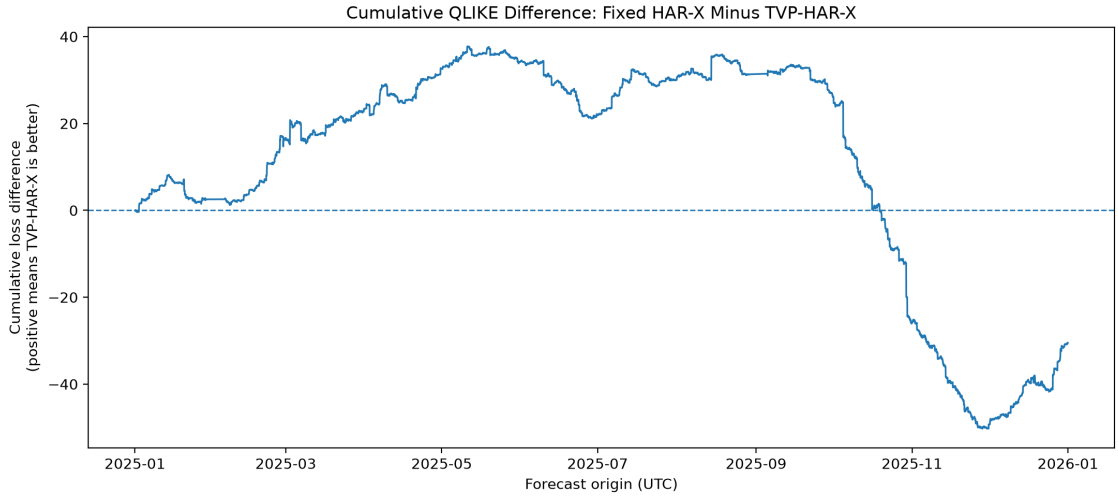


Figure 7: Cumulative QLIKE difference between Fixed HAR-X and TVP-HAR-X

Figure 7 displays the cumulative difference between Fixed HAR-X and TVP-HAR-X QLIKE. An upward segment indicates lower loss for TVP-HAR-X, while a downward segment indicates lower loss for Fixed HAR-X. The path confirms that neither specification maintains a uniform advantage throughout the test year.

5.11 Summary of Empirical Findings

The empirical results yield five main findings.

(1) Hourly log realized variance is persistent and mean reverting, although global level stationarity is rejected.

(2) AR(24) improves substantially on persistence, indicating that predictive dependence extends across several recent hours.

(3) Fixed HAR-X improves on AR(24), particularly under test QLIKE, supporting the use of multi-horizon volatility and market-activity predictors.

(4) Trading activity provides the strongest incremental predictor among the additional HAR-X variables.

(5) TVP-HAR-X achieves the lowest MAE and RMSE, while Fixed HAR-X achieves the lowest QLIKE. The gains from parameter variation therefore depend on the evaluation criterion.

The strongest and most stable improvement arises from moving from persistence and AR(24) to the HAR-X information set. Time-varying coefficients provide additional gains in log-scale forecast accuracy, while the fixed specification performs best under the primary variance-scale loss.

6 Discussion

6.1 Interpretation of the Main Result

The final test results produce different rankings across the three loss functions. TVP-HAR-X achieves the lowest MAE and RMSE on the log realized-variance scale, whereas Fixed HAR-X achieves the lowest QLIKE on the original variance scale.

MAE and RMSE evaluate errors in $y_t = \log(RV_t)$. MAE summarizes the typical absolute deviation, while RMSE assigns greater weight to large log-scale errors.

QLIKE evaluates the implied variance forecast $\widehat{RV}_t = \exp(\widehat{y}_t)$. Its loss depends on the ratio $RV_t/\widehat{RV}_t$, making it particularly sensitive to substantial underprediction during high-volatility periods (Patton, 2011). A model may therefore reduce average log-scale errors while recording a higher QLIKE because of a small number of unfavorable variance forecasts.

The test results suggest that coefficient adaptation improves the typical accuracy of log-volatility forecasts. Fixed HAR-X, however, provides slightly better performance under the variance-scale loss. The empirical value of parameter variation consequently depends on the forecasting objective.

6.2 Validation and Test Performance

TVP-HAR-X reduces validation QLIKE by approximately 4.30% relative to Fixed HAR-X, but its test QLIKE is 0.76% higher. This change in ranking illustrates the importance of retaining a separate final test period.

Fixed HAR-X estimates a single coefficient vector from the development sample and produces relatively stable forecasts. TVP-HAR-X introduces gradual coefficient movement, reducing bias when predictive relationships change while adding state-estimation uncertainty.

The relative performance of the two specifications can be understood as a trade-off between adaptation and estimation uncertainty. During validation, parameter variation improves all three losses. During the test period, it continues to reduce MAE and RMSE, but the gain is insufficient to lower QLIKE.

The difference across years also indicates that forecast-model rankings are sensitive to market conditions. Chronological validation is therefore more informative than a random split for this application.

6.3 Incremental Value of the HAR-X Information Set

The most substantial model improvement occurs between AR(24) and Fixed HAR-X. On the final test sample, Fixed HAR-X reduces QLIKE by approximately 5.75% relative to AR(24).

AR(24) is a demanding benchmark because it estimates a separate coefficient for each of the previous 24 hourly volatility observations. The HAR-X improvement therefore provides evidence that the structured predictor set contains information beyond an unrestricted one-day lag specification.

The HAR variables extend the information set to the preceding week without requiring 168 individual lag coefficients. This multi-horizon representation follows the HAR framework of Corsi (2009) and provides a parsimonious description of persistent volatility.

HAR-X also incorporates current market activity. Among the additional variables, log trade count displays the strongest and most stable association with next-hour volatility. The forecasting information set thus combines recent volatility, longer-horizon volatility conditions, and current trading activity.

6.4 Interpretation of Trading Activity

Trading activity is the most informative additional predictor in both HAR-X specifications. In the fixed model, its standardized coefficient is 0.3379, with a HAC p-value below 0.001. During validation, its time-varying coefficient remains positive and ranges from approximately 0.441 to 0.651.

These estimates indicate that unusually active trading hours tend to be followed by higher next-hour log realized variance, conditional on volatility measured at the four HAR horizons.

The result is consistent with the interpretation of trade count as a proxy for information arrival and market attention. It is related to the mixture-of-distributions framework of Clark (1973) and the transaction-count evidence reported by Jones et al. (1994). Increased activity may reflect periods in which market participants are processing new information and adjusting positions more intensively.

The estimated coefficient represents a conditional predictive association. Trading activity and volatility may respond jointly to latent information or broader changes in market conditions, so the model does not identify a causal effect.

6.5 Downside Semivariance

Downside semivariance is statistically insignificant in the fixed HAR-X model, and its time-varying coefficient changes sign during validation.

Realized semivariance separates the contributions of negative and positive intraday returns to total variation (Barndorff-Nielsen et al., 2010). In the present specification, total realized variance and the current-hour HAR component already contain the squared magnitudes of both positive and negative returns. Downside semivariance must therefore provide information beyond the overall volatility level to obtain a distinct effect.

The estimates suggest that this additional contribution is small on average at the one-hour forecast horizon. The logarithmic transformation may also reduce the influence of a limited number of extreme downside observations. The time-varying estimates do not reveal a persistent positive or negative relationship.

6.6 Aggressive-Buying Proxy

The aggressive-buying proxy has a coefficient close to zero in the fixed model and changes sign in the TVP specification.

The signed measure records the balance between taker-initiated buying and selling. Large imbalances in either direction may coincide with elevated volatility, which can weaken the average linear effect of a signed variable.

An absolute or nonlinear imbalance measure could capture this possibility more directly. Such alternatives were not included in the pre-specified model and would require evaluation on a new holdout sample.

The weak result may also reflect the information available in candlestick data. Aggregated taker-buy volume does not capture order-book depth, bid–ask spreads, cancellations, or trade-level price impact.

6.7 Time-Varying Coefficient Patterns

The state estimates reveal substantial differences in coefficient stability across predictors.

The six-hour and twenty-four-hour volatility coefficients remain positive and vary within relatively narrow ranges, indicating stable short- and medium-horizon persistence during validation.

The weekly coefficient fluctuates around zero, suggesting that the incremental contribution of longer-horizon volatility changes across market conditions. The downside-semivariance and aggressive-buying coefficients also cross zero and display limited average effects.

Trading activity remains positive throughout validation, although its magnitude changes over time. Its predictive contribution is therefore persistent within this period but not constant.

The state paths provide evidence that some components of the forecasting relationship are more stable than others. Short- and medium-horizon volatility persistence is relatively stable, while weekly and directional effects display greater variation. These patterns supplement the forecast comparison by showing how the conditional relationships evolve over time.

6.8 Stationarity and Parameter Variation

The ADF and KPSS results provide complementary evidence on the behavior of hourly log realized variance. The ADF test rejects a unit root, supporting the mean-reverting structure captured by AR(24). The KPSS test rejects global level stationarity (Kwiatkowski et al., 1992), indicating that the level or dependence structure may vary over longer periods.

This evidence is consistent with previous findings that structural breaks affect Bitcoin volatility forecasting (Shen et al., 2020). It also provides a statistical motivation for allowing gradual coefficient variation in the HAR-X relationship.

TVP-HAR-X retains the mean-reverting dependence represented by the HAR components while permitting their coefficients to evolve. The state-space specification therefore accommodates parameter instability within an otherwise persistent volatility process.

6.9 Residual Dependence

Ljung–Box portmanteau tests reject residual white noise for AR(24), Fixed HAR-X, and TVP-HAR-X (Ljung and Box, 1978). None of the specifications fully captures the temporal dependence in hourly volatility.

The remaining correlation may reflect long-memory behavior, intraday seasonality, nonlinear responses, omitted market variables, or changes in conditional variance. However, first-order residual and innovation autocorrelations are close to zero. For TVP-HAR-X, the lag-one autocorrelation of the standardized innovations is approximately 0.0007.

Given the large diagnostic samples, small correlations distributed across several lags can produce statistically significant portmanteau statistics. The residual diagnostics therefore indicate model incompleteness, while the out-of-sample results remain informative for relative forecast comparison.

6.10 Statistical and Practical Significance

The HAC loss-differential tests show that TVP-HAR-X reduces absolute and squared error relative to Fixed HAR-X. The improvements are 1.33% for absolute error and 1.87% for squared error, with one-sided p-values below 0.001.

These gains are statistically precise but modest in magnitude. Their practical relevance depends on the application. Small improvements may accumulate when forecasts are generated hourly and used repeatedly for risk limits, margin calculations, or position sizing. In settings where implementation simplicity is more important, the additional estimation and monitoring required by the state-space model may outweigh the reduction in average error.

Fixed HAR-X is simpler to estimate and deploy, while TVP-HAR-X provides lower log-scale forecast errors and information about changes in predictor importance. Model choice therefore depends on the relevant loss function and the value assigned to adaptability.

6.11 Empirical Design

Several features of the design strengthen the forecast comparison. The sample is divided chronologically, and all preprocessing parameters are estimated from the training period. Model selection is completed using validation data before the final test sample is evaluated.

All models are compared on identical forecast timestamps. State-space forecasts are generated sequentially, with each realized outcome entering the filter only after the corresponding forecast has been produced.

Fixed HAR-X and TVP-HAR-X use the same predictor set, so their performance difference reflects coefficient variation rather than unequal observable information. Forecast-loss comparisons also use HAC standard errors to account for serial dependence and heteroskedasticity.

These procedures reduce the scope for data leakage and ensure that the reported loss differences arise from the model specifications evaluated under common conditions.

6.12 Limitations

6.12.1 Single Market and Contract

The analysis is based on the Binance BTCUSDT USD-margined perpetual futures contract. The results may differ for Bitcoin spot markets, other exchanges, alternative cryptocurrency assets, or less liquid contracts. Exchange-specific trading mechanisms and participant composition may affect both market activity and volatility dynamics.

6.12.2 Single Forecast Horizon

The study considers only one-hour-ahead volatility. Predictor importance and model rankings may change at shorter or longer horizons. Order-flow variables may be more informative over several minutes, while longer-horizon volatility components may contribute more to multi-hour or daily forecasts.

6.12.3 Limited Microstructure Information

The available data include candlestick volume, trade count, and taker-buy volume, but exclude the complete limit order book. Bid–ask spreads, market depth, cancellations, liquidations, price impact, open interest, and funding conditions are not observed directly.

The market microstructure variables used here should therefore be interpreted as aggregate high-frequency trading proxies.

6.12.4 Linear Predictor Effects

Both HAR-X specifications impose linear effects in the observation equation. The relationship between activity and future volatility may be nonlinear, particularly during extreme market conditions.

A signed aggressive-buying measure may also understate the effect of order-flow imbalance when both strong buying and strong selling are associated with elevated volatility. Nonlinear and absolute-imbalance terms would require pre-specification and evaluation on a new holdout sample.

6.12.5 Restricted State Covariance

All seven time-varying slope coefficients share the common state innovation variance q_β. This restriction improves parsimony and numerical stability but imposes a common prior rate of variation across predictor effects.

Separate state variances could allow trading activity, short-horizon volatility, and weekly volatility coefficients to evolve at different rates. Such a specification would add hyperparameters and may increase estimation uncertainty.

6.12.6 Retransformation from Log Variance

The models forecast log realized variance, and the variance forecast used in QLIKE is obtained as

$$\widehat{RV}_t = \exp\left(\widehat{y}_t\right).$$

In general,

$$E\left[RV_t \mid \mathcal{F}_{t-1}\right] \neq \exp\left(E\left[\log RV_t \mid \mathcal{F}_{t-1}\right]\right).$$

The direct exponential transformation is closer to a conditional median than to a conditional mean when log forecast errors have positive variance. The same transformation is applied to every model, preserving internal comparability. Future work could evaluate lognormal or nonparametric bias corrections for variance-scale forecasting.

6.12.7 Seasonality and Long-Memory Structure

The feature set contains six-hour, daily, and weekly volatility averages but no explicit hour-of-day or day-of-week indicators. Continuous cryptocurrency trading may still display seasonal patterns associated with participation across geographic regions.

The significant long-horizon Ljung–Box statistics also suggest that additional seasonal or long-memory dependence remains.

6.13 Directions for Further Research

The analysis can be extended in several directions. Applying the same design to multiple contracts and exchanges would show whether the predictive value of trading activity generalizes across markets. Evaluating several forecast horizons would clarify how the importance of HAR components and activity variables changes with the prediction interval.

Richer high-frequency predictors could include order-book depth, bid–ask spreads, liquidation volume, open interest, and funding rates. Nonlinear transformations of trade count and order-flow imbalance may also capture effects missed by the linear specification.

Alternative state dynamics provide another extension. Mean-reverting coefficients or regime-switching models may be appropriate when forecasting relationships alternate among recurring market states. Forecast combinations could also exploit the relative stability of Fixed HAR-X and the adaptive behavior of TVP-HAR-X.

Any extension informed by the present test results would require a new validation period and an untouched final holdout sample.

7 Conclusion

This study evaluates one-hour-ahead Bitcoin volatility forecasts using five-minute BTCUSDT USD-margined perpetual futures data from 2022 to 2025. Hourly realized variance is forecast using a persistence benchmark, AR(24), Fixed HAR-X, and TVP-HAR-X within a chronological training–validation–test design.

The results show that multi-horizon volatility and current market activity provide useful predictive information beyond both persistence and a conventional autoregression. On the 2025 test sample, Fixed HAR-X reduces QLIKE from 0.5054 for AR(24) to 0.4764, an improvement of approximately 5.75%. Trading activity contributes the strongest incremental signal among the additional HAR-X predictors and remains positively associated with next-hour volatility throughout the validation period.

Allowing coefficients to vary over time produces a more qualified result. TVP-HAR-X achieves the lowest test MAE and RMSE, at 0.5901 and 0.7734, respectively. Relative to Fixed HAR-X, these values correspond to improvements of approximately 1.33% in absolute error and 1.87% in squared error. Both differences are statistically significant under the HAC loss-differential tests.

Fixed HAR-X nevertheless records the lowest test QLIKE, at 0.4764, compared with 0.4800 for TVP-HAR-X. The one-sided test provides no evidence that parameter variation improves the pre-specified primary loss. Coefficient adaptation therefore improves average log-scale accuracy but does not deliver a uniform advantage under variance-scale evaluation.

The evidence supports two main conclusions. First, short-horizon Bitcoin volatility is meaningfully predictable when recent volatility is represented across several horizons and combined with trading activity. Second, the value of time-varying coefficients depends on the forecast objective and market period. Fixed HAR-X provides the strongest performance under QLIKE, while TVP-HAR-X offers modest gains in MAE and RMSE and additional information about changes in predictor importance.

These findings are specific to one contract, exchange, and forecast horizon. Broader conclusions require evaluation across additional markets, horizons, and richer microstructure data.

Acknowledgements

The author used ChatGPT, developed by OpenAI, as an auxiliary tool during the preparation of this report. In particular, ChatGPT assisted with the preparation of the Abstract and Appendix, as well as the initial development of the Python scripts whose filenames begin with `download`, `extract`, and `inspect`. The remaining report content was written independently by the author, with AI-generated suggestions used only as references or sources of inspiration. All empirical results, numerical values, model interpretations, and final conclusions were checked by the author, who takes full responsibility for the content of the report. ChatGPT was also used to improve the language and clarity of the main text of this paper.

References

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625.

Barndorff-Nielsen, O. E., Kinnebrock, S., and Shephard, N. (2010). Measuring downside risk: Realised semivariance. In Bollerslev, T., Russell, J. R., and Watson, M. W., editors, *Volatility and Time Series Econometrics: Essays in Honor of Robert F. Engle*, pages 117–136. Oxford University Press, Oxford.

Bergsli, L. Ø., Lind, A. F., Molnár, P., and Polasik, M. (2022). Forecasting volatility of bitcoin. *Research in International Business and Finance*, 59:101540.

Binance (n.d.). Binance public data. `https://github.com/binance/binance-public-data`. Accessed July 15, 2026.

Clark, P. K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1):135–155.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Durbin, J. and Koopman, S. J. (2012). *Time Series Analysis by State Space Methods*. Oxford University Press, Oxford, 2 edition.

Jones, C. M., Kaul, G., and Lipson, M. L. (1994). Transactions, volume, and volatility. *The Review of Financial Studies*, 7(4):631–651.

Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45.

Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., and Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root: How sure are we that economic time series have a unit root? *Journal of Econometrics*, 54(1–3):159–178.

Ljung, G. M. and Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2):297–303.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256.

Said, S. E. and Dickey, D. A. (1984). Testing for unit roots in autoregressive-moving average models of unknown order. *Biometrika*, 71(3):599–607.

Shen, D., Urquhart, A., and Wang, P. (2020). Forecasting the volatility of bitcoin: The importance of jumps and structural breaks. *European Financial Management*, 26(5):1294–1323.

A Reproducibility and Additional Results

B Reproducibility

B.1 Software Environment

The empirical analysis was conducted in Python 3.11.15 using the Conda environment `btc-volatility`. The complete dependency specification is provided in `environment.yml`.

The principal packages are `pandas` and `numpy` for data processing and numerical computation, `pyarrow` for Parquet input and output, `statsmodels` for regression estimation, HAC inference, and time-series diagnostics, `scikit-learn` for supporting preprocessing operations, and `matplotlib` for figure generation. The data-acquisition scripts also use `requests` and `tqdm`.

The environment can be reconstructed from the project root using

```
conda env create -f environment.yml
conda activate btc-volatility
```

B.2 Execution Procedure

The report is accompanied by a code archive containing the source files, environment specification, and a `README.md` file. The README documents the Binance data source, required directory structure, complete execution order, and expected output files. The original data can be retrieved from the public source using the included download and extraction scripts.

After data acquisition and preprocessing, the principal analysis scripts are executed from the project root in the following order:

```
python src/evaluate_naive_baseline.py
python src/run_complete_ar_analysis.py
python src/run_fixed_harx_analysis.py
python src/run_tvp_harx_analysis.py
python src/run_final_test_evaluation.py
python src/create_final_report_outputs.py
```

These scripts respectively generate the persistence forecasts, select and evaluate the autoregressive model, estimate the fixed HAR-X model, estimate the TVP-HAR-X state-space model, perform the locked final test evaluation, and produce the tables and figures used in the report.

Scripts whose filenames begin with `download`, `extract`, and `inspect` handle data acquisition, archive extraction, and preliminary data-quality inspection. Detailed commands for these preprocessing stages are provided in `README.md`.

B.3 Reproducibility Controls

All timestamps are processed in UTC, and lagged variables are constructed on the complete hourly calendar. Transformation and standardization parameters are estimated from the training sample and then held fixed.

Model and hyperparameter selection uses the validation period. The 2025 test sample is evaluated only after the model specifications have been fixed. All models are compared on the same set of forecast timestamps, and the state-space forecasts are generated sequentially without using future observations.

Generated numerical results, tables, and figures are written to the `outputs` directory. Running the documented pipeline reconstructs the empirical outputs reported in the main text.